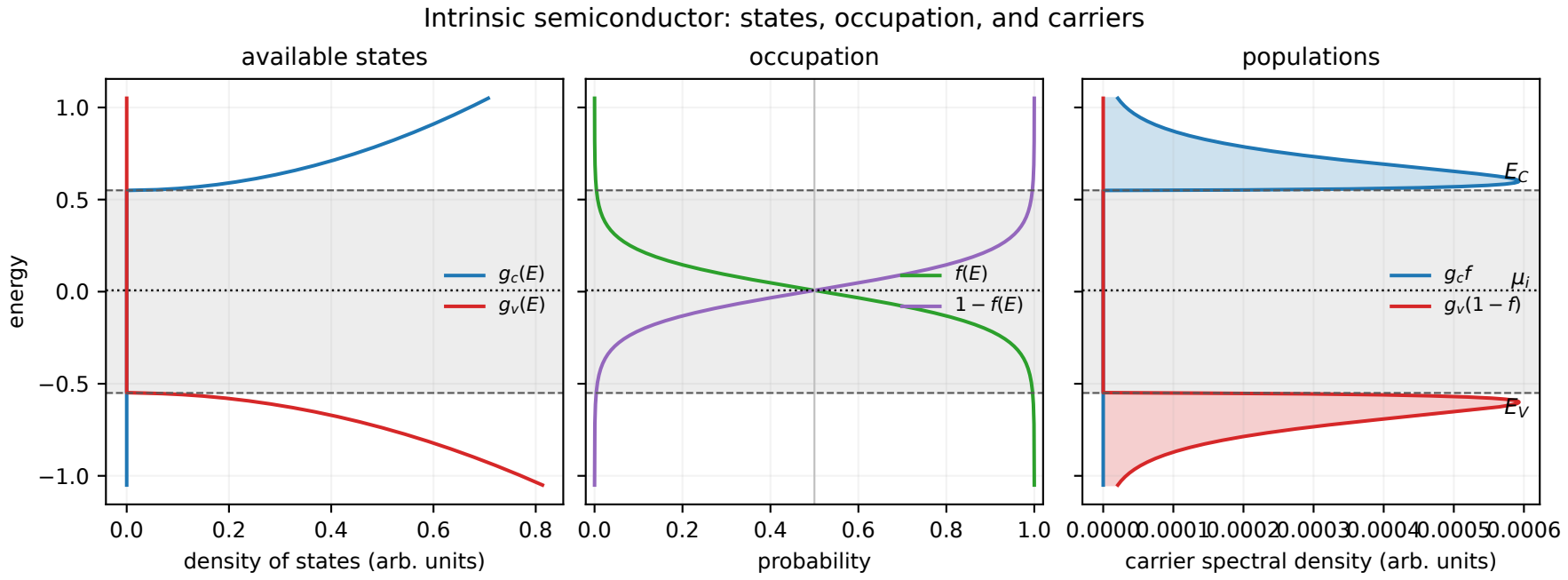

Intrinsic semiconductor: states, occupation, and carriers
available states
occupation
populations
energy
1.0
0.5
0.0
−0.5
−1.0
$g_c(E)$
$g_v(E)$
$f(E)$
$1-f(E)$
E_C
E_V
μ_i
$g_c f$
$g_v(1-f)$
0.0
0.2
0.4
0.6
0.8
1.0
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
0.0006
density of states (arb. units)
probability
carrier spectral density (arb. units)